TrustCharge

AI-Based Centralized EV Charging Reliability and Smart Reservation Platform

GROUP 9, S7C

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Project Abstract

TrustCharge is a centralized platform designed to unify fragmented Electric Vehicle (EV) charging networks into a single, seamless ecosystem. The project innovatively integrates multiple charging providers to provide a cohesive user experience. By utilizing advanced Artificial Intelligence models, the system predicts charger health, assesses functional reliability, and calculates accurate queue waiting times. The core goal of the platform is to eliminate range anxiety and provide data-driven recommendations that effectively streamline the charging process for all users. .

Introduction & Importance

Rising Demand & Struggles

- The rapid adoption of electric vehicles has drastically increased the demand for charging infrastructure.
- Users often struggle to identify reliable stations, resulting in unnecessary travel and range anxiety.
- Unpredictable station queues lead to severe and frustrating waiting times.

The Centralized Solution

- ✓ Provides accurate information and intelligent recommendations to improve the charging experience.
- ✓ EV & Fleet Owners: Find reliable stations quickly and efficiently manage schedules.
- ✓ Station Operators: Increase station utilization rates and boost satisfaction.
- ✓ Encourages wider EV adoption and directly supports sustainable transportation globally.

Problem Statement

To address the unreliability of fragmented EV charging networks, there is a critical need for a platform that centralizes real-time availability data from different providers and integrates a unified reservation system, thereby eliminating long waiting times and severe range anxiety.

Existing Systems vs. Drawbacks

Existing Systems

Applications like Tata Power EZ Charge, Statiq, ChargeZone, and Bolt.Earth provide basic functionalities:

- Locating nearby charging stations via map.
- Basic navigation features to reach the station.
- In-app payment facilities for charging.

However, each app works independently and serves only its own network.

Major Drawbacks

- ✕ Separate applications needed for different providers.
- ✕ No centralized membership or wallet system.
- ✕ Complete absence of AI-based reliability prediction (users arrive at broken chargers).
- ✕ Limited queue prediction capabilities.
- ✕ Users cannot effectively compare stations across networks.

The Proposed System

A Centralized Hub

TrustCharge introduces a centralized platform connecting multiple charging providers into a single application utilizing AI and real-time data.

AI Reliability Score & Charger Health

Predicts if a charger is actually functioning before arrival.

Unified Wallet

A single membership system for all networks.

Cross-Net Reserve

Book slots in advance across providers.

AI Queue Predict

Highly accurate waiting time estimation.

Functional Requirements

- **User Account & Membership Management:** The system shall provide secure user authentication and manage a unified membership account accessible across all supported charging networks.
- **Centralized Station Discovery:** The platform shall integrate data from OpenChargeMap and OpenStreetMap to display comprehensive, real-time information about charging stations from multiple providers in one unified interface.
- **AI-Driven Reliability & Queue Prediction:** The system shall utilize AI models to analyze historical and real-time data to predict individual charger health/reliability and provide accurate queue waiting time estimations to the user.
- **Smart Reservation System:** The application shall allow users to perform prepaid slot bookings and manage reservations across different charging networks.
- **Real-time Navigation & Routing:** The system shall provide turn-by-turn navigation to recommended charging stations based on the user's current location and battery status.
- **Priority Access Management:** The platform shall facilitate membership-based priority access to ensure specific users can secure charging slots during peak demand.

Non-Functional Requirements

01

Scalability

The platform must be highly scalable to seamlessly accommodate new charging providers and expanding geographic regions without performance degradation.

02

Performance & Latency

AI predictions, route recommendations, and real-time station availability updates must be processed and delivered to the user rapidly with minimal latency.

03

Security

Implementation of robust security protocols and encryption for the unified wallet system, secure reservations, and end-to-end protection of user data.

Software Requirements

Frontend

- React.js for user interface
- Tailwind CSS for responsive design

Backend

- Node.js for asynchronous operations
- Express for API endpoint management
- MySQL for relational database storage

AI & Authentication

- Firebase Authentication for secure identity management
- AI-based models for charger health prediction
- AI-based queue waiting time estimation
- OpenChargeMap and OpenStreetMap APIs for location/navigation data

Hardware Requirements

Processor: Any standard modern processor (e.g., Intel Core i3 or equivalent)
Memory (RAM): 4 GB minimum
Storage: 50 GB of available disk space
Network: Standard internet connection for API access

Future Scope

Global Expansion

Scaling the platform's footprint across multiple states and integrating international networks.

V2G Integration

Incorporating smart grid features to allow EV owners to sell energy back during peak hours.

Autonomous Fleets

Developing API endpoints for self-driving vehicle fleets to autonomously route and reserve slots.

Dynamic Pricing

AI-driven cost recommendations helping users schedule charges when grid rates are lowest.

A Smarter Way to Charge

TrustCharge provides a centralized, AI-powered EV charging solution that improves reliability and significantly reduces waiting times.

It offers a seamless charging experience across different charging networks through intelligent route planning and unified membership.

THANK YOU